

Experiment 6

**AIM: Create Product Catalog Application using ASP.NET MVC Architecture.
(Controllers, Models, Views, Routing)**

Source: Controller

Code: ProductDefaultController.cs

```
using
System.Collections.Gen
eric; using
System.Web.Mvc;
using Practical_06.Models;

namespace Practical_06.Controllers
{
    public class ProductDefaultController : Controller
    {
        public ActionResult Index()
        {
            IList<Product> products = GetProducts();

            return View(products);
        }

        public ActionResult Details(int id)
        {
            IList<Product> products = GetProducts();

            Product product = products.Find(p => p.ProductId == id);

            if (product == null)
            {
                return HttpNotFound();
            }

            return View(product);
        }

        private IList<Product> GetProducts()
        {
            return new IList<Product>()

```

```
{
  new Product
  {
    ProductId = 1, ProductName =
      "laptop",
      Description = "HP laptop-Powerful laptop with an Intel Core i5
        processor,fast performance,and ample storage.",
    Price = 55000,
    Category = "Electronics",
    Quantity = 10,
    Image = "https://encrypted-
      tbn0.gstatic.com/images?q=tbn:ANd9GcSZ9sHs5lsqtdz1ujCp
      GRvI-VvV3dV_c85Flwj0gSZFXQsvHVbVP1AwT3Mks=10"
  },

  new Product
  {
    ProductId = 2,
    ProductName = "iPhone 18 Pro Max",
    Description = "A premium Apple smartphone featuring a powerful A20 Pro
      chip, advanced camera system, and a large 6.9-inch display.",
    Price = 150000, Category =
      "Electronics", Quantity =
      15,
    Image = "https://images.news18.com/ibnlive/uploads/2026/07/iphone-18-
      pro-max-dark-cherry-colour-2026-07-a3cb605328de47234af80743a3fe4428-
      e1785468879176-1200x675.png?impolicy=websitekwidth=400kheight=225"
  },

  new Product
  {
    ProductId = 3,
    ProductName = "Headphones",
    Description = "Wireless Bluetooth Headphones - Premium headphones with
      clear sound, deep bass, and comfortable ear cushions.",
    Price = 2500,
    Category = "Accessories",
    Quantity = 20,
    Image = "https://images.unsplash.com/photo-
      1546435770-a3e426bf472b?w=400"
  }
}
```

```
}  
}  
}
```

Source: Model Code:

Product.cs using

System;

```
        namespace Practical_06.Models  
{  
    public class Product  
    {  
        public int ProductId { get; set; }  
  
        public string ProductName { get;  
set; } public string Description {  
get; set; } public decimal Price {  
get; set; } public string Category  
{ get; set; } public int Quantity {  
get; set; }  
  
        public string Image { get; set; }  
    }  
}
```

Source: Views/ProductDefault

Code: Details.cshtml

```
        @model Practical_06.Models.Product  
  
@{  
    layout = null;  
}  
  
    <!DOCTYPE html>
```

```
<html>
<head>
<meta charset="utf-8" />
<title>Product Details</title>
</head>

<body style="font-family:Arial;
background-color:#f5f5f5;
margin:0;">

<!-- Left Side Container -->
<div style="width:650px;
margin:40px 0 40px 40px;">

<!-- Heading -->
<h1 style="font-
size:32px;
color:#222;
margin-bottom:25px;">
Product Details
</h1>

<!-- Details Table -->
<table border="1"
cellpadding="15"
cellspacing="0"
width="100%"
style="border-
collapse:collapse;
background-color:white;
font-size:18px;
color:#222;">

<!-- Product Image -->
<tr>
<td colspan="2"
style="text-
align:center;
padding:20px;">



</td>
</tr>

<!-- Product ID -->
<tr>
  <th style="width:35%;
    text-align:left;
    background-color:white;">
    Product ID
  </th>

  <td>
    <b>@Model.ProductId</b>
  </td>
</tr>

<!-- Product Name -->
<tr>
  <th style="text-align:left;
    background-color:white;">
    Product Name
  </th>

  <td>
    <b>@Model.ProductName</b>
  </td>
</tr>

<!-- Description -->
<tr>
  <th style="text-align:left;
    background-color:white;">
    Description
  </th>

  <td>
    @Model.Description
  </td>
</tr>
```

```
<!-- Price -->
<tr>
  <th style="text-align:left;
    background-color:white;">
    Price
  </th>

  <td>
    <b>₹ @Model.Price</b>
  </td>
</tr>

<!-- Category -->
<tr>
  <th style="text-align:left;
    background-color:white;">
    Category
  </th>

  <td>
    @Model.Category
  </td>
</tr>

<!-- Quantity -->
<tr>
  <th style="text-align:left;
    background-color:white;">
    Quantity
  </th>

  <td>
    @Model.Quantity
  </td>
</tr>

</table>

<!-- Back link -->
<div style="margin-top:20px;">

  <a href="@Url.Action("Index", "ProductDefault")">
```

```
style="font-size:18px;
color:#222;
font-weight:bold;
text-decoration:none;">
```

← Back to Product Catalog

```
<
/
a
>
```

```
<
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>
```

```
</div>
```

```
</body>
</html>
```

Code: index.cshtml

```
@model
```

```
IEnumerable<Practical_06.Models.Produc
```

```
t> @{
layout = null;
ViewBag.Title = "Product Catalog";
}
```

```
<!DOCTYPE html>
<html>
<head>
<meta charset="utf-8" />
<title>Product Catalog</title>
</head>
```

```
<body style="font-family:Arial; background-color:#f5f5f5; margin:0;">
```

```
<div style="width:98%; margin:30px auto;">
```

```
<!-- Heading -->
<h1 style="text-align:center; font-size:32px;
color:#222;"> Product Catalog
```

</h1>

<p style="text-align:center; font-size:19px; color: 222; font-weight:bold;">

Available Products

</p>

<!-- Product Table -->

```
<table border="1"
  cellpadding="15"
  cellspacing="0"
  width="100%"
  style="border-
    collapse:collapse;
    background-color:white;
    font-size: 17px;
    color:#222;">
```

<!-- Header -->

```
<tr style="background-color:#333;
  color:white;
  text-
  align:center;
  font-
  size:18px;">
```

```
<th style="width:15%;">Product Image</th>
<th style="width:8%;">Product ID</th>
<th style="width:15%;">Product Name</th>
<th style="width:22%;">Description</th>
<th style="width:10%;">Price</th>
<th style="width:12%;">Category</th>
<th style="width:8%;">Quantity</th>
<th style="width:10%;">Action</th>
```

</tr>

<!-- Products -->

```
@foreach (var product in Model)
{
```

```
  <tr
    style="text-
    align:center;
    vertical-
    align:middle;
    color:#222;">
```

<!-- Image -->

```
<td>
  
    </td>

    <!-- Product ID -->
    <td>
        <b>@product.ProductId</b>
    </td>

    <!-- Product Name -->
    <td>
        <b style="font-
        size:19px;">
            @product.ProductName
        </b>
    </td>

    <!-- Description -->
    <td style="line-height:1.5; font-weight:bold;">
        @product.Description
    </td>

    <!-- Price -->
    <td>
        <b style="font-size:18px;">
            ₹ @product.Price
        </b>
    </td>

    <!-- Category -->
    <td>
        <b>@product.Category</b>
    </td>

    <!-- Quantity -->
    <td>
        <b>@product.Quantity</b>
    </td>

    <!-- Action -->
    <td>
        <a href="@Url.Action("Details", "ProductDefault", new { id = product.ProductId

```

```

})
"
        style="display:inline-
        block; padding:10px
        15px; background-
        color:#333;
        color:white;
        text-
        decoration:none;
        border-radius:5px;
        font-weight:bold;">
        View Details
    </a>
</td>

</tr>
}

</table>

</div>

</body>
</html>

```

Source: Shared Code:

_Layout.cshtml

```

<!DOCTYPE html>
<html>
<head>
<meta charset="utf-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>@ViewBag.Title - My ASP.NET Application</title>
<link href="~/Content/Site.css" rel="stylesheet" type="text/css" />
<link href="~/Content/bootstrap.min.css" rel="stylesheet" type="text/css" />
<script src="~/Scripts/modernizr-2.6.2.js"></script>
</head>
<body>
<div class="navbar navbar-inverse navbar-fixed-top">
    <div class="container">
        <div class="navbar-header">
            <button type="button" class="navbar-toggle" data-toggle="collapse" data-
            tar_get=".navbar-collapse">

```

```

        <span class="icon-bar"></span>
        <span class="icon-bar"></span>
        <span class="icon-bar"></span>
    </button>
        @Html.ActionLink("Application name", "Index", "Home", new { area = ""
            }, new { @class = "navbar-brand" })
    </div>
    <div class="navbar-collapse collapse">
        <ul class="nav navbar-nav">
        </ul>
    </div>
</div>
</div>

<div class="container body-content">
    @RenderBody()
    <hr />
    <footer>
        <p>kcopy; @DateTime.Now.Year - My ASP.NET Application</p>
    </footer>
</div>

<script src="~/Scripts/jquery-1.10.2.min.js"></script>
<script src="~/Scripts/bootstrap.min.js"></script>
    </body>
</html>

```

Source: App_Start

Code: RouteConfig.cs

```

        using
        System.Web.Mvc
        c; using
        System.Web.Routing;

        namespace Practical_06
    {
        public class RouteConfig
        {
            public static void RegisterRoutes(RouteCollection routes)
            {
                routes.IgnoreRoute("{resource}.axd/{*pathInfo}");
            }
        }
    }

```

```
routes.MapMvcAttributeRoutes();

routes.MapRoute( name:
    "Default",
    url: "{controller}/{action}/{id}",
    defaults: new
    {
        controller = "ProductDefault",
        action = "Index",
        id = UrlParameter.Optional
    }
);
}
```

Output: Screenshot:

Product Catalog
Available Products

Product Image	Product ID	Product Name	Description	Price	Category	Quantity	Action
	1	Laptop	HP Laptop—Powerful laptop with an Intel Core i5 processor, fast performance, and ample storage.	₹ 55000	Electronics	10	View Details
	2	iPhone 18 Pro Max	A premium Apple smartphone featuring a powerful A20 Pro chip, advanced camera system, and a large 6.9-inch display.	₹ 150000	Electronics	15	View Details
	3	Headphones	Wireless Bluetooth Headphones – Premium headphones with clear sound, deep bass, and comfortable ear cushions.	₹ 2500	Accessories	20	View Details

Product Details



Product ID	1
Product Name	Laptop
Description	HP Laptop-Powerful laptop with an Intel Core i5 processor,fast performance,and ample storage.
Price	₹ 55000
Category	Electronics
Quantity	10

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Product Details



Product ID	2
Product Name	iPhone 18 Pro Max
Description	A premium Apple smartphone featuring a powerful A20 Pro chip, advanced camera system, and a large 6.9-inch display.
Price	₹ 150000
Category	Electronics
Quantity	15

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Product Details



Product ID	3
Product Name	Headphones
Description	Wireless Bluetooth Headphones – Premium headphones with clear sound, deep bass, and comfortable ear cushions.
Price	₹ 2500
Category	Accessories
Quantity	20

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Github Link:

<https://github.com/Rajyaguruparthraj/.netlab1/tree/main>